

Cookies in PHP

What is a Cookie?

A cookie is a **small piece of information stored inside the user's browser**.

It helps the website remember the user.

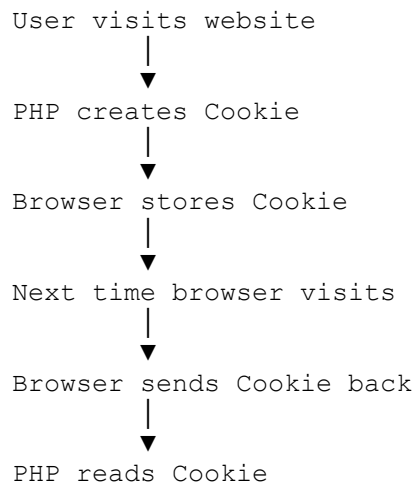
Examples:

- Remember username
- Dark mode preference
- Language selection
- Shopping cart

Think of it as:

"Hello Browser! Keep this information until I ask for it."

How Cookies Work



Syntax of setcookie()

```
setcookie(name, value, expiry_time, path);
```

Example

```
setcookie("username", "Rakesh", time()+3600);
```

Here,

- username → Cookie name
- Rakesh → Value
- time()+3600 → Expires after 1 hour

Important Rule

setcookie() must be called before any HTML output.

✗ Wrong

```
echo "Hello";

setcookie("user", "Rakesh", time()+3600);
```

✓ Correct

```
<?php
setcookie("user", "Rakesh", time()+3600);
?>

<html>
```

Creating a Cookie

```
<?php

setcookie("username", "Rakesh", time()+3600);

echo "Cookie Created";

?>
```

Output

Cookie Created

Browser stores

username = Rakesh

Accessing Cookie

Cookies are accessed using

```
$_COOKIE
```

Example

```
<?php  
  
echo $_COOKIE["username"];  
  
?>
```

Output

Rakesh

Checking Cookie Exists

```
<?php  
  
if(isset($_COOKIE["username"]))  
{  
    echo "Welcome ".$_COOKIE["username"];  
}  
else  
{  
    echo "Cookie not found";  
}  
  
?>
```

Output

Welcome Rakesh

Deleting Cookie

Set expiry time in the past.

```
setcookie("username","",time()-3600);
```

Browser deletes it.

Complete Example

Create cookie

```
<?php  
  
setcookie("city","Surat",time()+3600);  
  
?>
```

Read cookie

```
<?php  
  
echo $_COOKIE["city"];  
  
?>
```

Output

Surat

Real-life Uses of Cookies

- Remember Me
- Shopping Cart
- Theme
- Language
- Login Preference

Advantages

- ✓ Fast
- ✓ Stored in browser
- ✓ Easy to use

Disadvantages

- ✗ User can delete cookies
- ✗ Limited storage (about 4 KB)
- ✗ Less secure

Session Management

What is Session?

A session stores information **on the server**, not in the browser.

Imagine

School Office keeps attendance.

Students don't carry attendance register.

That is Session.

Difference

Cookie

Stored in Browser

Session

Stored on Server

Session Workflow

User Login



Server creates Session ID



Browser stores Session ID



Actual Data stays on Server

Starting Session

Before using session,

```
session_start();
```

must be called.

Like

Open attendance register first.

Syntax

```
session_start();
```

Creating Session

```
<?php  
  
session_start();  
  
$_SESSION["username"]="Rakesh";  
  
echo "Session Created";  
  
?>
```

Access Session

```
<?php  
  
session_start();  
  
echo $_SESSION["username"];  
  
?>
```

Output

Rakesh

Checking Session

```
<?php

session_start();

if(isset($_SESSION["username"]))
{
    echo "Welcome ".$_SESSION["username"];
}
else
{
    echo "Login First";
}

?>
```

Destroy Session

```
<?php

session_start();

session_destroy();

?>
```

Everything stored inside session is removed.

Remove One Session Variable

```
unset($_SESSION["username"]);
```

Login Example

login.php

```
<?php

session_start();

$_SESSION["username"]="Admin";

echo "Login Successful";

?>
```

home.php

```
<?php
session_start();
echo "Welcome ".$_SESSION["username"];
?>
```

Output

Welcome Admin

Advantages of Session

- ✓ More Secure
- ✓ Stored on Server
- ✓ Used for Login
- ✓ No size limitation like cookies

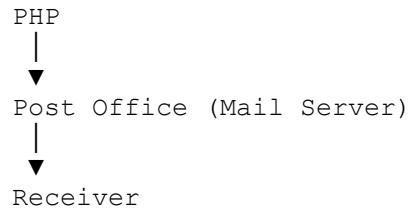
Session vs Cookie

Cookie	Session
Stored in Browser	Stored on Server
Less Secure	More Secure
Small Storage	Larger Storage
Can be Deleted Easily	Server Controls It
Used for Preferences	Used for Login

Sending Emails using mail()

PHP can send emails.

Imagine



Syntax

```
mail(to, subject, message);
```

Example

```
<?php
mail(
    "abc@gmail.com",
    "Test Email",
    "Hello from PHP"
);
echo "Mail Sent";
?>
```

Complete Syntax

```
mail(
    to,
    subject,
    message,
    headers
);
```

Example

```
<?php
$to="abc@gmail.com";

$subject="Welcome";

$message="Welcome to our Website";

$headers="From: admin@college.com";

mail($to,$subject,$message,$headers);

?>
```

Important

The `mail()` function works **only if the server is configured with a mail service** (such as Sendmail on Linux or SMTP through a configured server). On many local environments like XAMPP, it **does not work without additional configuration**.

For production applications, developers often use libraries like **PHPMailer** or SMTP services because they are more reliable.

Email Formatting

Emails look professional because of headers.

What are Headers?

Headers tell email programs

- Who sent it
- HTML or Plain Text
- CC
- BCC
- Reply To

Example

```
$headers="From: admin@gmail.com";
```

Multiple Headers

```
$headers="From: admin@gmail.com\r\n";  
$headers.="Reply-To: support@gmail.com\r\n";  
$headers.="CC: teacher@gmail.com\r\n";  
$headers.="BCC: manager@gmail.com";
```

Sending HTML Email

Without header

```
<h1>Hello</h1>
```

appears as text.

With header

It becomes

Hello

Example

```
<?php  
$to="abc@gmail.com";  
$subject="Welcome";  
$message="<h1>Welcome Student</h1>";  
$headers="MIME-Version: 1.0\r\n";  
$headers.="Content-type:text/html;charset=UTF-8\r\n";  
$headers.="From: admin@gmail.com";  
mail($to,$subject,$message,$headers);  
?>
```

Email Subject

Subject appears at the top.

Example

Subject:
Welcome to PHP Course

In PHP

```
$subject="Welcome to PHP Course";
```

Complete Email Example

```
<?php

$to = "student@example.com";

$subject = "PHP Practical";

$message = "<h2>Hello Student</h2>";
$message .= "<p>Your practical has been submitted successfully.</p>";

$headers = "MIME-Version: 1.0\r\n";
$headers .= "Content-type:text/html;charset=UTF-8\r\n";
$headers .= "From: admin@college.com\r\n";

if(mail($to, $subject, $message, $headers))
{
    echo "Email Sent Successfully";
}
else
{
    echo "Email Sending Failed";
}

?>
```

Summary Table

Topic	Function	Purpose
Create Cookie	<code>setcookie()</code>	Store small data in the browser
Read Cookie	<code>\$_COOKIE</code>	Access cookie values
Start Session	<code>session_start()</code>	Begin a session
Store Session	<code>\$_SESSION</code>	Store data on the server
Remove Session	<code>unset()</code> , <code>session_destroy()</code>	Delete session data
Send Email	<code>mail()</code>	Send an email (requires mail server configuration)
Email Subject	<code>\$subject</code>	Set the email title
Email Headers	<code>\$headers</code>	Define sender, HTML format, CC/BCC, etc.
Attachments	PHPMailer (recommended)	Send files with an email

(Viva)

1. **Cookie** stores data in the **browser** and is created using `setcookie()`.
2. **Session** stores data on the **server** and is started using `session_start()`.
3. Use `$_COOKIE` to read cookies and `$_SESSION` to access session variables.
4. `mail()` sends emails but requires a properly configured mail server.
5. Use **email headers** to specify sender information, HTML content type, CC/BCC, and other metadata.
6. For sending **attachments**, prefer **PHPMailer** instead of the basic `mail()` function.